

MSDS 7330

File Organization and Database Management In Class Assessment 8

- 1) What does 'NoSQL' mean today?
 - a) Not ever SQL
 - b) Not only SQL
 - c) No SQL
 - d) Never once SQL
- 2) What does the CAP theorem state?
 - a) A distributed database can optimize only one of C, A or P.
 - b) A single machine database can optimize all of C, A and P simultaneously.
 - c) A distributed database can optimize only two of C, A or P simultaneously.
 - d) A distributed database can optimize all three of C, A and P simultaneously.
- 3) What does the 'C' in CAP stand for?
 - a) Managing consistency
 - b) Managing compatibility
 - c) Managing computations
 - d) Managing calculations
- 4) What does the 'A' in CAP stand for?
 - a) Actionable
 - b) Arbitrary
 - c) Analysis
 - d) Availability
- 5) What does the 'P' in CAP stand for?
 - a) Performance
 - b) Parameterized data
 - c) Partition toleration
 - d) Probability
- 6) Which of the following is a type of document database?
 - a) MongoDB
 - b) Cassandra
 - c) Riak
 - d) Neo4j
- 7) Which of the following is a type of graph database?
 - a) MongoDB
 - b) Cassandra
 - c) Riak
 - d) Neo4j

8) Which of the following is a type of column-family database?

- a) MongoDB
- b) Cassandra
- c) Riak
- d) Neo4j

9) Which of the following is a type of key-value database?

- a) MongoDB
- b) Cassandra
- c) Riak
- d) Neo4j

10) Where does most of the power of the graph database

- a) the relationships and their properties
- b) the placement of the nodes in a three dimensional space
- c) the nodes and their properties
- d) the limited number of relationships between nodes

11) Which of the following is not a reason NoSQL has become a popular solution for some businesses?

- a) Better scalability than RDBMS.
- b) Improved ability to keep data consistent.
- c) Faster access to data than RDBMS.
- d) More easily allows for data to be held across multiple servers.

12) Why would you choose a NoSQL database over an RDBMS?

- a) To improve programmer productivity by using a database that better matches an application's needs.
- b) To improve data access performance.
- c) To improve system scalability to very large data sets.
- d) All of the above.
- e) None of the above.

13) Graph databases are:

- a) generally useful for storing session information, user profiles, preferences, shopping cart data. We would avoid using Key-value databases when we need to query by data, have relationships between the data being stored or we need to operate on multiple keys at the same time.
- b) generally useful for content management systems, blogging platforms, web analytics, real-time analytics, ecommerce-applications. We would avoid using document databases for systems that need complex transactions spanning multiple operations or queries against varying aggregate structures.
- c) generally useful for content management systems, blogging platforms, maintaining counters, expiring usage, heavy write volume such as log aggregation. We would avoid using column family databases for systems that are in early development, changing query patterns.
- d) very well suited to problem spaces where we have connected data, such as social networks, spatial data, routing information for goods and money, recommendation engines.

14) Document databases are:

- a) generally useful for storing session information, user profiles, preferences, shopping cart data. We would avoid using Key-value databases when we need to query by data, have relationships between the data being stored or we need to operate on multiple keys at the same time.
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15) Key-Value databases are:

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